Assignment

WII

- Back in 2008, hospitals were reporting an increase in injuries related to playing Nintendo Wii, mainly attributed to muscle and tendon strains.
- A researcher hypothesized that a stretching warm-up before playing Wii would help lower injuries, and the athletes would be less susceptible to injuries.
- She took 60 athletes and 60 non-athletes, half of them played Wii and half watched others playing as a control, and within these groups half did a 5-min stretch routing before playing/watching whereas the other half did not.
- The outcome was a pain score out of 10 (0=no pain, 10 is severe) after playing Wii for 4 hours.

Load: **Wii.dat**

- Conduct an appropriate ANOVA to test whether athletes are less prone to injury, and whether the prevention program worked.

Assignment

MUSICAL TASTES

- People's musical tastes change as they get older. This is a hypothesis to be tested.
- Two groups of people (young and old, n=45 in each group) were recruited. Then we split them into 3 groups and assign them to listen to different music types: Fugazi, ABBA or Barf Grooks.
- After listening to the music we got each person to rate it on a scale ranging from -100 (dislike) through 0 to +100 (like).

Load: **fugazi.dat**

- According to the design, please describe the specific hypotheses in your opinion.
- Please conduct an ANOVA test and appropriate post hoc tests to verify your hypotheses.

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Assignment

TEXT MESSAGE

- Text messaging is very popular among mobile phone owners. One concern is that children may use this form of communication so much that it will hinder their ability to learn correct written English.
- One researcher conducted an experiment in which one group of children (n=25) was encouraged to send text messages on mobile phones over a 6-month period and forbid the second group (n=25) from sending any text messages.
- The grammatical test (as percentage) was measured both before and after the experiment.

Load:
TextMessages.dat

- In this scenario, what kind of ANOVA should you use?
- Based on what you've learned from ANOVAs, please test whether text messaging affects English grammar among children.

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Assignment

CHECK THE PAPER

Synchrony Between Default-Mode and Sensorimotor Networks Facilitates Motor Function in Stroke Rehabilitation: A Pilot fMRI Study

Load:
Stroke_Rehab12.csv

Behavior Improvements After Stroke Rehabilitation

Considering their motor performances, the repeated-measure ANCOVA showed statistical significance of FMA (**Figure 1A**) and TEMPA (**Figure 1C**) scores along the rehabilitation process ($FMA_{Pre:Post:Follow}$ = 26.8:29.3:30.3, $F_{2,22}$ = 6.3, $p < 0.007$; $TEMPA_{Pre:Post:Follow}$ = −59.8:−55.3:−50.2, Huynh-Feldt-corrected $F_{1.4,15.5}$ = 7.9, $p < 0.008$), but WMFT (**Figure 1B**) did not show a significant difference ($WMFT_{Pre:Post:Follow}$ = 45.5:48.0:46.9, $F_{2,22}$ = 6.3, $p < 0.007$; $TEMPA_{Pre:Post:Follow}$ = 45.5:48.0:46.9, $F_{2,22}$ = 1.30, p = 0.294). The *post hoc* comparison demonstrated significant FMA enhancements after the rehabilitation process (Pre vs. Post and Follow, Bonferroni-corrected $p < 0.05$), and TEMPA was enhanced between Pre and Post states (Bonferroni-corrected p = 0.03).

- **Check the Results section in the attached paper.**
- **Try to replicate the findings in the first paragraph of Results.**
- **Given the dataset, please refine Fig.1 and generate a correlation matrix, highlighting the coefficients with significance.**

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